

Anthropogenic Carbon Sequestration and Deep Ocean Circulation Dynamics under Climate Change

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ABSTRACT

Marine ecosystems and thermohaline circulation play a crucial role in absorbing anthropogenic carbon dioxide emissions from the atmosphere. Using coupled ocean-atmosphere general circulation models, we evaluate deep ocean carbon uptake and ocean acidification trajectories. The slowdown of the Atlantic Meridional Overturning Circulation significantly impairs global oceanic carbon sequestration capacity.

Keywords: Climate Change, Carbon Sequestration, Ocean Acidification, Greenhouse Gas Emissions, Global Warming

1. Introduction

The oceans have sequestered approximately 30% of anthropogenic CO₂ emissions since the industrial revolution. However, oceanic carbon sink efficiency is governed by temperature stratification, biological pumps, and deep-water formation.

2. Climate Model Projections

We analyzed multi-century simulation scenarios under high emissions pathways. Surface warming weakens polar downwelling, suppressing the transport of dissolved inorganic carbon into deep abyssal layers.

3. Ecological Impacts

Decreasing pH levels accelerate ocean acidification, threatening calcifying marine organisms and reef ecosystems. The projected weakening of the global carbon sink will amplify atmospheric warming feedbacks.

4. Conclusion

Preserving oceanic carbon sequestration requires urgent global decarbonization and aggressive reduction in greenhouse gas emissions to prevent irreversible climatic tipping points.